RATING METHODOLOGY

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Rating Methodology

Medical Products and Devices

This rating methodology replaces the *Medical Products and Devices* methodology published on 26 October 2023. We have made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies. These updates do not change our methodological approach.

Scope

This rating methodology applies to companies globally that are primarily engaged in the sale of medical products and devices.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

Sector overview

Manufacturers of medical products and devices are a diverse group, united primarily by the customers — healthcare providers or research entities — they serve. Hospitals, a key customer base for companies in this sector, purchase technology-intensive devices, such as stents and implantable cardioverter-defibrillators, as well as more commodity-like products, such as syringes and catheters. Laboratories and academic centers, another key customer base, purchase equipment, instruments and consumables needed to conduct diagnostic testing or research.

In the US, the FDA oversees product approval, inspection of manufacturing facilities and ongoing product-specific safety matters. Outside of the US, medical device companies are subject to similar oversight by national regulators. The level of regulatory risk may vary depending on a company's specific product lines or its warning letters outstanding from the FDA or other regulators.

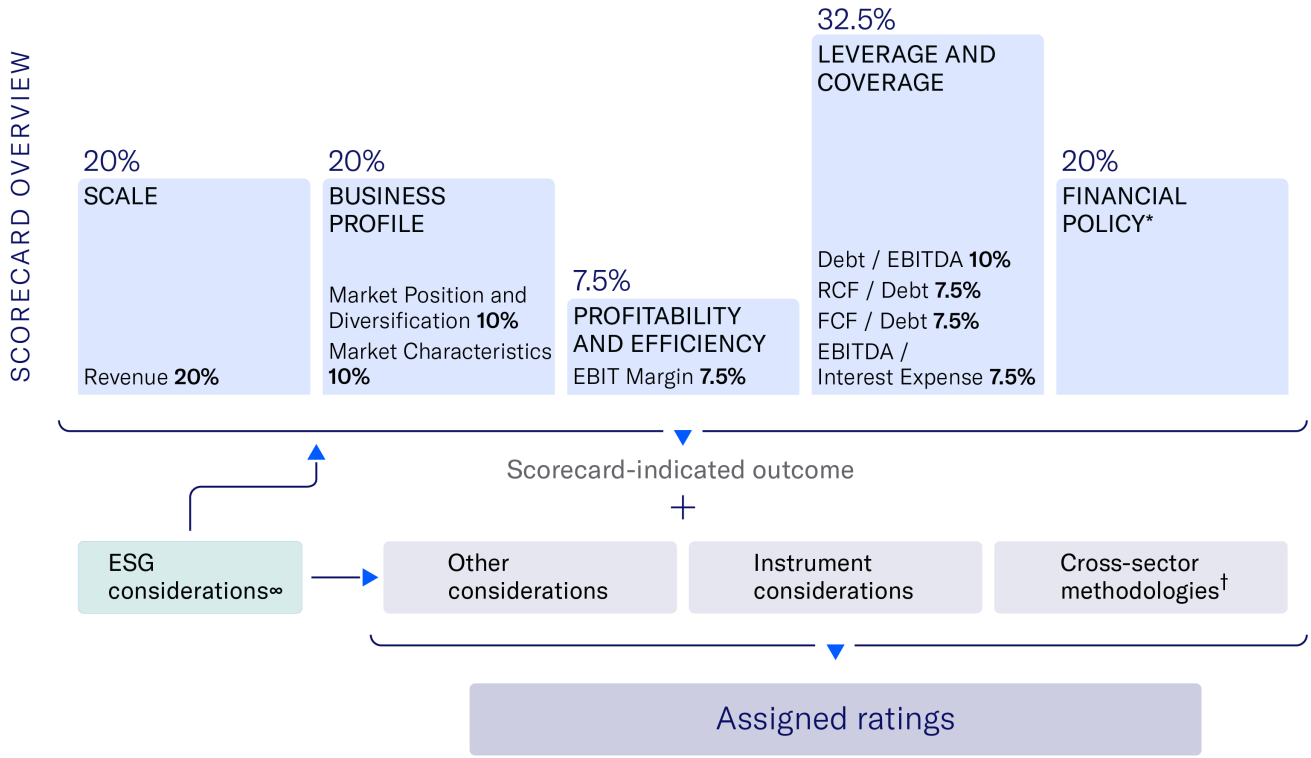
Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1

Illustration of the medical products and devices rating approach



* This factor has no subfactors.

∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.

† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Medical products and devices scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (20%)									
Revenue (USD Billion)[1]	20%	≥ $40	$20 - $40	$10 - $20	$4 - $10	$1.5 - $4	$0.5 - $1.5	$0.25 - $0.5	< $0.25
Factor: Business Profile (20%)									
Market Position and Diversification	10%	Extremely diverse group of broad product segments with distinct end-user markets. Estimated to have very limited reliance on any product segment and No. 1 market positions or >50% market share in majority of segments.	Highly diverse group of broad product segments with distinct end-user markets. Estimated to have limited reliance on any product segment and No. 1 or No. 2 market positions in majority of segments.	Well-diversified group of broad product segments with distinct end-user markets. Estimated to have some reliance on one product segment and No. 1 or No. 2 market positions in majority of segments.	Diverse group of broad product segments with distinct end-user markets. Estimated to have moderate reliance on one product segment. Among top three players in all significant segments.	Moderately diverse group of broad product segments with distinct end-user markets. Estimated to have moderately high reliance on one product segment. Among leading players in most segments.	One broad product segment or one primary end-user market. Estimated to have high reliance on one product segment. Among leading players in this segment.	At least one niche product segment with leading market presence. Estimated to have very high reliance on this niche segment.	Not expected to have a significant market presence in any product segment.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Business Profile (20%)									
Market Characteristics	10%	Exceptional prospects for organic sales growth and market share stability due to: (1) extremely high level of innovation and demand; (2) minimal pricing constraints across all geographic regions; (3) extremely low regulatory, technology or competitive risk.	Superior prospects for organic sales growth and market share stability due to: (1) very strong level of innovation or demand; (2) low pricing constraints across all geographic regions; (3) very low regulatory, technology or competitive risk.	Very good prospects for organic sales growth and market share stability due to: (1) strong level of innovation or demand; (2) moderately low pricing constraints across all geographic regions; (3) low regulatory, technology or competitive risk.	Good prospects for organic sales growth and market share stability due to: (1) above-average level of innovation or demand; (2) moderate pricing constraints across all geographic regions; (3) average regulatory, technology or competitive risk.	Moderate prospects for organic sales growth or market share stability due to: (1) average level of innovation or demand; (2) moderately strong pricing constraints across all geographic regions; (3) above-average regulatory, technology or competitive risk.	Modest prospects for organic sales growth or market share stability due to: (1) below-average level of innovation or demand; (2) strong pricing constraints across all geographic regions; (3) high regulatory, technology or competitive risk.	Expect flat to declining sales growth and weak market share stability due to: (1) very weak level of innovation or demand; (2) very strong pricing constraints across all geographic regions; (3) very high regulatory, technology or competitive risk.	Expect declining sales growth and loss of market share due to: (1) lack of innovation or demand; (2) severe pricing constraints across all geographic regions; (3) severe regulatory, technology or competitive risk.
Factor: Profitability and Efficiency (7.5%)									
EBIT Margin[2]	7.5%	≥ 40%	30% - 40%	20% - 30%	15% - 20%	10% - 15%	5% - 10%	2.5% - 5%	< 2.5%
Factor: Leverage and Coverage (32.5%)									
Debt / EBITDA[3]	10%	≤ 0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4.5x	4.5x - 6.5x	6.5x - 8.5x	> 8.5x
RCF / Debt[4]	7.5%	≥ 60%	45% - 60%	35% - 45%	25% - 35%	15% - 25%	5% - 15%	0% - 5%	< 0%
FCF / Debt[5]	7.5%	≥ 45%	30% - 45%	20% - 30%	10% - 20%	5% - 10%	0% - 5%	(5)% - 0%	< (5)%
EBITDA / Interest Expense[6]	7.5%	≥ 30x	20x - 30x	10x - 20x	7x - 10x	4x - 7x	2x - 4x	1x - 2x	< 1x

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Financial Policy (20%)									
Financial Policy	20%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests. Although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $60 billion. A value of $60 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 70%. A value of 70% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[4] For the linear scoring scale, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is (5)%. A value of (5)% or worse equates to a numeric score of 20.5.
[5] For the linear scoring scale, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is (10)%. A value of (10)% or worse equates to a numeric score of 20.5.
[6] For the linear scoring scale, the Aaa endpoint value is 40x. A value of 40x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[1] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and its bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (20% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

In this sector, a company's market presence and product diversity, as well as the characteristics of its products and the markets it serves, provide a meaningful indicator of the likely stability and sustainability of its future cash flow. Product line and end-user diversification help offset the constantly evolving dynamics of the healthcare delivery system, which influence demand and pricing. Market share can indicate competitiveness, depth of customer relationships and likely prospects for future performance.

Many companies in this sector manufacture a vast array of products, each with distinct characteristics and customer bases worldwide. The Business Profile factor includes an assessment of the types of products a company manufactures, as well as the company-specific and sector-wide variables that can influence prospects for sales growth and market share shifts.

How we assess it for the scorecard

We score this factor based on two qualitative subfactors: Market Position and Diversification, and Market Characteristics.

Market Position and Diversification

We score this subfactor based on our assessment of several dimensions of the company's business profile: (1) market positioning within product lines, which typically includes a company's breadth of reach; (2) diversity of product lines, including those outside of medical devices; (3) level of concentration or reliance on one or two segments; and (4) distinct customer or end-user categories, which are critical to purchasing decisions.

Companies often report product lines as "sales segments," which comprise products that treat a specific medical or clinical market or serve a specific healthcare customer's need. Examples include orthopedic products, dental products, eye care products, cardiovascular products, urology products, endosurgery products, hospital beds and clinical laboratory products. In assessing this subfactor, we consider the degree of reliance on one product line or product, especially in light of competition and potential regulatory issues.

Companies with similar broad product lines may offer different products. Breadth of product offerings within a product line or business segment can influence the scoring within this subfactor. For example, one orthopedic company may offer only spinal products while another may offer reconstructive, trauma and spinal products. In rare instances, a company may manufacture only one distinct product within one customer segment; such a company would typically have a low score for this subfactor.

We assess a company's market share in its major product lines. The breadth and depth of product positions often influence our forward view of market share. For certain devices, particularly those for which sales representatives are critical in helping physicians use the products, relationships tend to be more "sticky," meaning customers do not change vendors easily. Fewer device products are considered "sticky" than in the past, as hospitals have become more influential in making decisions about purchases. We view favorably companies that are highly diverse from a product line or end-user standpoint and that hold leading (No. 1 or No. 2) market positions in most of these product lines.

We typically view high customer and end-user diversification positively when it mitigates the risk that a change in an individual end-user market (such as cuts in hospital reimbursement or funding for academic research by the US National Institutes of Health or a similar organization) will significantly impair profitability and cash flow. Healthcare organizations such as hospitals, clinical laboratories, pharmaceutical companies and academic research centers, and physicians within a specialty area, are examples of customer bases or end-user markets.

Certain product lines may have multiple targeted customer bases. For example, clinical laboratory products may be sold to both hospital laboratories and reference laboratories. Manufacturers of dental products may serve both dentists and dental laboratories. Likewise, multiple product lines may have a single customer base. For example, hospitals are the primary purchasers of infusion pumps and hospital beds.

We typically view lack of concentration in any product line or lack of reliance on one end-user category favorably. Even with a relatively high level of diversification in product offerings, concentration in one or two product lines or reliance on one end-user category can negatively influence scoring for this subfactor.

Market Characteristics

For this subfactor, we assess the company's prospects for organic sales growth and market share stability. We may consider the following in our assessment of market characteristics: (1) product life cycle and the risk of technological obsolescence; (2) strength of demand for products across geographies; (3) pricing constraints across geographies; (4) number of and quality of competitors; (5) risk that regulatory matters, including safety or efficacy concerns, will impede sales growth; (6) level of innovation, including strength of pipeline; and (7) ability to demonstrate value to key decision makers, including hospitals and payors.

Product life cycle and the risk of technological obsolescence are important components of this subfactor. Companies that make products that are more highly technology-oriented tend to face challenges related to shorter life cycles, including higher obsolescence risk and more stringent pre-market and post-market regulatory oversight. For example, manufacturers of certain cardiac devices face risks related to the potential for customers to rapidly adopt a competing new technology, which would engender shifts in market share

and cash flow. The rapid pace of change forces many medical device companies to invest more in R&D. Manufacturers of products with longer life cycles (such as hospital beds or syringes) require less R&D investment and generally have more stable market positions because of lower risk that a new technology will render their products outdated. Companies that make products with longer life cycles typically also have less exposure to regulatory issues.

All medical device companies need to innovate in order to offset pricing pressure with new value-added products. We consider the quality of a company's pipeline and its success, not only with product approvals, but also in demonstrating value to its key customers.

Another component that can be company- or sector-specific is demand. Company-specific regulatory matters such as recall actions or Food and Drug Administration (FDA) warning letters, i.e., a notification that a manufacturer has significantly violated FDA regulations, or industrywide questions related to safety or the appropriate utilization of a device can negatively affect demand and sales.

Pricing trends are also important, especially if customers are under reimbursement pressure or if a highly competitive market forces prices down. Because most companies sell products globally, demand and pricing trends can vary across geographies depending on the maturity of the products in each region, including the level of product penetration, demographic differences, clinical practice patterns, differences in reimbursement and coverage and payor constraints.

Factor: Profitability and Efficiency (7.5% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

For companies in this sector, profits are also needed to maintain sufficient reinvestment in research and development. Highly technical devices may provide significant medical value and contribute greatly to improving a patient's health. Healthcare payors will often allow for additional reimbursement of these devices, and hospitals may exert less pricing pressure on the medical device maker if the product can be shown to significantly improve patient outcomes, including reducing complications or hospital readmissions. As a result, some of these devices can be richly priced and have very high operating margins. Manufacturers of low-technology-oriented, more-commodity-like products may have less favorable pricing and, consequently, lower margins. Given the lower value-added nature of these products, efficient operations may be even more critical to ensure price competitiveness.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBIT Margin.[2]

EBIT Margin

We use the ratio of earnings before interest and taxes (EBIT) to revenue.

Factor: Leverage and Coverage (32.5% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on four quantitative subfactors: Debt/EBITDA; RCF/Debt; FCF/Debt; and EBITDA/Interest Expense.[3]

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

RCF / Debt

We use the ratio of retained cash flow (RCF) to total debt.

FCF / Debt

We use the ratio of free cash flow (FCF) to total debt.

EBITDA / Interest Expense

We use the ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) to interest expense.

Factor: Financial Policy (20% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

Many companies in this sector have historically used acquisitions to spur revenue growth, expand business lines, consolidate market positions, advance cost synergies or seek access to new technology.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[4]

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[5]

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

In this sector, regulatory considerations also play a role in our assessment of a company's market characteristics. A material, negative change in a company's regulatory status or large damages resulting from litigation may, however, cause ratings to be lower than a scorecard-indicated outcome based on historical financials. This is especially the case if a regulatory change is swift or the uninsured portion of a liability is very large.

Medical device companies are also exposed to legal challenges, including product liability or patent lawsuits. Safety concerns highlighted by FDA recalls, for example, often result in product liability claims. The presence of these legal matters presents risk, because they could result in cash outlays, lower levels of cash and higher leverage. We typically view the outlays associated with legal matters as one-time in nature.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary

and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic

incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[6]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[7]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Geographic diversity

The majority of rated companies in this sector operate globally. As a result, geographic diversity does not generally play an important role in the scorecard. Nonetheless, we consider geographic diversity beneficial because it helps mitigate the effects of economic downturns, unfavorable regulations or negative trends in healthcare delivery. Instances of significant geographic concentration would likely be captured in the Business Profile factor and more generally incorporated into our ratings.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4
Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[9]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[10]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[11] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

11 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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